

## GRAINGUARD · DESIGN DECISION RECORD

# GrainGuard recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | GGD-DDR-002                                                   | Version | 0.4            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-10-02     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/grainguard                       |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                          |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| 0.4     | 2026-10-02 | Amish Chadha | 01, 02 and 04 decided by Amish on 2026-10-02 as recommended in GGD-DEC-001                                                      |
| 0.3     | 2026-09-27 | Amish Chadha | Solar panel faces away from the bin: decided by Amish on 2026-09-27; model.py rotation sign flipped, drawing GGD-DWG-001 Rev P3 |
| 0.2     | 2026-09-26 | Amish Chadha | Budget set to \$260 to cover the priced BOM: decided by Amish, 2026-09-26; R13 met                                              |
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's acceptance of all recommendations and the changes made in the repo                                               |

## 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation (O1, O2 and O4) were given recommendations in the design decisions register (GGD-DEC-001) and decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions." The budget was decided by Amish on 2026-09-26 (see Budget, 2026-09-26). The solar panel facing was decided by Amish on 2026-09-27 (see Solar panel facing, 2026-09-27).

### ■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." After GGD-DDR-001, GrainGuard had one open item with a recommendation (O3, the plenum probe), two component specifications set in GGD-CAL-001 v0.1 and flagged for Amish (a relay input of 3 mA or less and a charger clamped at 15.0 V), two open items with no recommendation (O1 and O2) and one question on the pitch wording with options but no recommendation. Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and the repo stays at TRL 3.

### ■ Decision

*Table 1. Items decided by Amish, 2026-09-25: go with recommendation.*

| #                 | Item                                                                                                                                                                                                                                                 | Decision                                                 | What changed in the repo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1<br>to<br>D11   | TRL 2 review items (GGD-DDR-001): budget option (a), pitch reword, control rule, RH sensing, one center cable, RS-485, solar controller with relay kit, AGM, plenum sensing if affordable, CO <sub>2</sub> as an option, corn in a small Midwest bin | Already decided in GGD-DDR-001; confirmed                | No change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| N1<br>(was<br>O3) | Adopt the plenum temperature probe despite the budget                                                                                                                                                                                                | Adopt it into the base kit and accept the \$6.50 overrun | <code>bom/bom.csv</code> item 14 is no longer an option. Per-bin kit \$247.00 to \$256.50 (2.6 % over \$250); first prototype with receiver \$269.00 to \$278.50. <code>budget_usd</code> stays \$250, and R13 records the accepted overrun (status "not met, overrun accepted"), following the recommendation to accept rather than raise the budget. GGD-CAL-001 v0.2 adds check B10: with a DS18B20 ( $\pm 0.5$ °C) and the ambient SHT45 ( $\pm 0.1$ °C), the fan heat is known within $\pm 0.51$ °C and the plenum EMC within about $\pm 0.50$ points, against a 1.85-point spread when the fan heat is only a setting. R4 moves from at risk to met on paper and is restated in GGD-REQ-001 v0.4. GGD-PRC-001 v0.4, drawing GGD-DWG-001 Rev P2 (notes), concept media (labels, key figures, flow) and <code>cad/src/model.py</code> comment updated; STEP and STL re-exported (geometry unchanged, the probe was already modeled) |
| N2                | Relay input of 3 mA or less (flagged in GGD-CAL-001 v0.1)                                                                                                                                                                                            | Keep as specified                                        | No change; recorded in GGD-PRC-001 v0.4, Key design choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| N3                | Charger compensation clamped at 15.0 V (flagged in GGD-CAL-001 v0.1)                                                                                                                                                                                 | Keep as specified                                        | No change; recorded in GGD-PRC-001 v0.4, Key design choices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Table 2. Items left open by this record, all decided on 2026-10-02.

| #  | Item                                                                                                                                                                                                                                               | Status                                                                                                                                                                                                                                                                                                                                                     |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | First host farm and electrician for a co-design visit. No partner was recommended.                                                                                                                                                                 | <b>Decided by Amish, 2026-10-02:</b> recruit the host farm and its electrician through a land-grant extension grain-storage specialist, for example at Purdue University or Iowa State University, choosing a farm with a small corn bin, an aeration fan and a center hanger; the route to a first candidate, not an agreed partner (GGD-DEC-001, item 4) |
| 02 | Battery and panel capacity for the recovery half of R8: a 20 W panel (about \$8 to \$10, further over budget), relax the recovery target to about 7 days, or accept the miss. No recommendation was made.                                          | <b>Decided by Amish, 2026-10-02:</b> a 20 W panel, provided the mast check with the stay removed is rerun for the larger panel and still passes (GGD-DEC-001, item 2)                                                                                                                                                                                      |
| 04 | Pitch wording against the cool-mode rewet bound (at most 74 kg of water per cycle, one point in the bottom 0.37 m): keep "without rewetting it", say only "cool it or dry it", or cut the cool-mode allowance to zero. No recommendation was made. | <b>Decided by Amish, 2026-10-02:</b> say only "cool it or dry it" in the pitch and keep the cool-mode allowance (GGD-DEC-001, item 3)                                                                                                                                                                                                                      |

Table 3. Decided but on hold.

| Item                                                                                                                                                                                                                         | Status                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| TRL 4 work named in the review note (bench pod and plenum probe with a lab test report, membrane equilibration time, fan heat on a real fan, relay kit fail-off timing, cold-soak of battery and charger, build log entries) | Decided as the next step, on hold: TRL 4 is on hold by Amish's instruction |

No cross-repo action arises: GrainGuard does not use SwapCell or any other portfolio module.

### Budget, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items." Budget set to \$260 to cover the priced BOM: decided by Amish, 2026-09-26. The priced per-bin kit is \$256.50 ( `bom/bom.csv` , items 1 to 11, 13 and 14, with the farmhouse receiver costed per farm under GGD-DDR-001 D1), so `budget_usd` in `project.yaml` moves from 250 to 260. This supersedes the N1 choice to keep \$250 and record the overrun: R13 moves from "not met, overrun accepted" to met. `docs/04-calcs/sizing.py` reads the budget from `project.yaml` and was rerun; GGD-CAL-001 v0.3, GGD-REQ-001 v0.5, GGD-PRC-001 v0.5, GGD-PRB-001 v0.4, `README.md` and `bom/bom-notes.md` were updated. Requirement status is now met 8, at risk 3, not met 1 (R8) and not verifiable 2.

### Solar panel facing, 2026-09-27

On 2026-09-27 Amish wrote: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense." For GrainGuard the challenge was item 2 of the appearance-model review (`docs/REVIEW.md`, session 2026-09-26): `cad/src/model.py` rotated the solar panel envelope with `Rot(0, -panel_tilt, 0)` , which tilts its face toward the bin, although its comment says "facing away from the bin"; 0.7 m from a 5.6 m wall the panel would then sit in the wall's shade. The recommendation was to change the rotation to `Rot(0, panel_tilt, 0)` and regenerate the media. Decided by Amish on 2026-09-27: go with the recommendation.

- `cad/src/model.py` : the panel rotation is now `Rot(0, panel_tilt, 0)` , so the face tilts 45 degrees from horizontal toward the outside of the bin (local +X). Panel size, tilt angle, height and position on the mast are unchanged.

- `cad/src/product_model.py` : the appearance model already faced the panel away from the bin with the same rotation. Its comment that `model.py` "rotates its envelope box the other way" was removed; its pose now matches `model.py` with no compensation.
- Regenerated: STEP and STL ( `cad/step` , `cad/stl` ), drawing GGD-DWG-001 Rev P3 (detail B and the views show the panel facing outward) and the concept media ( `media/hero.png` , `concept-blueprint` , `cutaway.png` , `exploded.png` , `flow.png` , `model.glb` ).
- GGD-PRC-001 v0.6 states the facing in its component table. No requirement status changes: the R8 energy figures in GGD-CAL-001 already assumed a panel in sun, and the wind load uses the panel area, which is unchanged.

## ■ Consequences

- Requirement status (GGD-CAL-001 v0.2): not met 1 (R8); not met with an accepted overrun 1 (R13); at risk 3 (R2, R6, R9); not verifiable at TRL 3 2 (R3, R10); met 7 (R1, R4, R5, R7, R11, R12, R14). Before this record: not met 1, at risk 4, not verifiable 2, met 7.
- Controlled documents bumped: GGD-PRC-001 v0.4, GGD-REQ-001 v0.4, GGD-CAL-001 v0.2, GGD-DDR-001 v0.2; drawing GGD-DWG-001 Rev P2.
- `project.yaml` : `budget_usd` stayed 250 under N1 (set to 260 on 2026-09-26, see above), pitch and problem unchanged, `trl`: 3 and `trl_target`: 3 unchanged.